

Representation Convergence · Complete Experimental Summary

Testing the Platonic Representation Hypothesis: do independently trained encoders converge to one structure? All numbers verified against the final report.

1 · The encoders · architecture, how each was pretrained, and its role

encoder / dim	architecture	objective + what supervises it	supervision regime	role and measured result
DINOV2 s/b/l 768/1536/2048	ViT (transformer)	Self-distillation (DINO image + iBOT patch); EMA-teacher targets on other crops of the same image; no text	SELF-SUPERVISED (auto-curated data)	Hub members + capacity ladder: 40.9/53.5/58.3% of ceiling; +15.3 pts/decade, fit R² 0.949. img_small (768) sets hub width.
bge-m3 1024	XLM-R (transformer)	Contrastive on mined query-passage pairs; the pairing is the label	PAIR-SUPERVISED (text only)	Hub member + the head's TARGET. Half of the top pair with SBERT (raw Procrustes +0.738).
GPT-2 768	Transformer (causal)	Next-token prediction; hold-out-and-predict on tokens	SELF-SUPERVISED	The collapse case: eff.rank 6.6, readable 95.1% / writable 24.8%. Whitening lifts R@1 0.130-0.478 = 102.6% of a contrastive encoder (C.11).
BERT 768	Transformer (masked)	Masked language modelling	SELF-SUPERVISED	Hub member, ladder rung. eff.rank 65.0 · between GPT-2 and the contrastive text spaces.
SBERT 768	BERT + NLI head	BERT fine-tuned on NLI entailment pairs (human-labelled)	SUPERVISED (strict sense)	Best head target: 101.0% of native. Healthiest text space (pair-cos +0.135). Other half of the top pair.
SigLIP 2 768	ViT (transformer)	Image-text contrastive on web alt-text; sees text	PAIR-SUPERVISED (HELD OUT)	Cross-lineage transfer 94.2%, inside the 93.8-96.5 band. Reproduced to the decimal by G0-exact.
ConvNeXt-base 1024	CONVNET (no attention)	ImageNet-22k classification on 22k human categories	FULLY SUPERVISED (HELD OUT)	96.7% of native · ABOVE the band and above SigLIP in the same run. Architecture does not bound the claim.
MobileCLIP-S1 SigLIP 2 (ExpB)	ViT (both)	Contrastive (DataCompDR / web alt-text)	PAIR-SUPERVISED	Experiment B pair: phone-server adapter. cos 0.900, R@1 0.987, AUC 0.999.
Qwen3-Emb 0.6B/4B 1024/2560	Qwen3 (causal, FT for embed)	Causal LM fine-tuned for embeddings; last-token pooling	CONTRASTIVE FT	Scale-extension text ladder. · 0.352 / 0.307 · the falsification of P3.
DINOV2-giant 3072 (cls+patch)	ViT (transformer)	Self-distillation (as DINOv2 ladder)	SELF-SUPERVISED	Scale-extension top rung. · 0.339 < large's 0.418 · the falsification of P2. Cannot be a hub member (-=1.26).

2 · The experiments · what each tested, how, and what it found

experiment	question	method	measurements used	result
A WikiText	Do two text encoders share structure on the same passages?	10,000 WikiText passages through two encoders; fit one linear map	R ² , cosine-to-target, R@1	Measurable but weak; robust to caption protocol (run both ways, same band).
B MobileCLIP-SigLIP	Can a phone-tier encoder be mapped into a server-tier space with one matrix?	4,000 COCO images; ridge + Procrustes adapters on 3k/1k split; Core ML export	held-out R ² , cosine, R@1/5, verification AUC	cos 0.900, R@1 0.987, AUC 0.999 (0.848 hardest), 88.4% of ceiling. 0.79 MB shipped. Linear IS the answer (MLP gain +0.0000).
C / vec2vec unsupervised	Recover the correspondence from SHAPE alone, no pairs?	Gromov-Wasserstein on intra-space distance matrices + CSLS self-learning	exact-match vs 1/N chance, retrieval R@1, verification AUC	33.5% exact match (vs 0.067% chance). Whitening DESTROYS it (50.5%-0.5%) · the sharpest negative.
E-series ladder	Does raw agreement order by isotropy, not 'language understanding'?	7 encoders, pairwise; effective rank, whitening rescue, pre-registered	effective rank, pair-cosine, raw R@1, whitened R@1	GPT-2 6.6 collapsed · whitening 102.6% of contrastive. Isotropy is what similarity training supplies.
G-series the hub	Can ONE frozen shared space serve seven encoders?	512-d whitened PCA of 7 concatenated spaces; one linear map per encoder; one frozen head	transfer AUC, R@1, % of native, verification AUC, kNN overlap	AUC 0.986, R@1 0.451, 42/42 pairs at 103.6%. Head trained on ONE encoder reads four others at 93-96%.
G4 held-out	Does shared DINOv2 ancestry explain the transfer?	SigLIP + ConvNeXt (no DINOv2 lineage) mapped into the FROZEN hub	% of native, alignment shuffle test, calibration control	SigLIP 94.2%, ConvNeXt 96.7% · a convnet ABOVE the band. Ancestry was not doing the work.
G11-G16 width cliff	Why does transfer collapse past hub width 768?	Sweep hub width; test 8 accounts against the curve; pre-register locations	transfer R@1 vs width, singular-value spectrum	SEVEN accounts falsified, the eighth (RANK WALL) confirmed to single-direction resolution: collapse at the source's ambient dimension.
Scale extension P1-P4	Does cross-modal alignment strengthen with encoder capacity?	Add DINOv2-giant + Qwen-0.6B/4B; 11 encoders, 55 pairs; pre-registered · thresholds	Spearman · (raw shape), hub transfer, McNemar noise test	P2 & P3 FALSIFIED: both ladders rise then FALL (image 0.418-0.339, text 0.352-0.307). Alignment not monotonic in capacity (0.02-4B).

3 · The measurements, and the headline numbers

measurement	what it asks	chance / reference
raw cross-space cosine	Do two encoders place the same item in the same DIRECTION?	· 0.00 (high-dim random vectors are near-orthogonal)
Spearman · of pairwise distances	Do the two spaces agree on which items are FAR from which? (no map fitted)	the PRH alignment measure; independent of any hub
kNN overlap (k=10)	Do the two spaces share nearest neighbours?	chance 0.010; measured 0.62 = 62× (local agreement)
R@1 / R@5 (retrieval)	Does the correct item rank first / in top 5?	chance 0.001 = 1/1000 gallery
% of native head	How much of a purpose-fit head does a FROZEN head recover?	100% = frozen head loses nothing
verification AUC	Can one cosine threshold separate a true pair from a random one?	0.5 = chance; the weaker question of D.2
effective rank (uncentred)	How many directions carry real variance?	full = healthy; GPT-2 = 6.6 (collapsed)
mean pair-cosine	Is one shared direction crowding the space? (anisotropy)	· 0 healthy; +0.999 = collapsed (GPT-2)

Headline results

Convergence is REAL but LOCAL. Encoders agree on neighbourhoods (kNN 62× chance, · up to 0.76) while raw coordinates are orthogonal (cos·0). Different charts of one manifold, not one space.

The strongest evidence is a FAILURE. Seven explanations of the width cliff were pre-registered and falsified; the eighth (rank wall) confirmed. A test that could have killed the account, and didn't.

Objective > lineage > modality. bge·SBERT (0.76, shared OBJECTIVE, no lineage) beats img·img (0.56, shared ViT-g ancestry). A shared training goal aligns more than shared architecture.

Capacity does NOT monotonically help. Both ladders rise then fall (0.02·4B): image 0.418·0.339 at giant, text 0.352·0.307 at Qwen-4B. Pre-registered, both falsified. (70B regime untested.)

Isotropy is a FRAME, not content. Whitening a collapsed space (GPT-2) rescues retrieval (0.130·0.478) but destroys unsupervised matching (50.5%·0.5%). Same transform, opposite effect.

4 · The cat / dog intuition · one worked example of experiment, measurement, result, and caveat

The setup. Imagine a tiny 2-D embedding of animal photos: cats cluster on one side, dogs on the other, and the whole cloud is stretched · wide in one direction, thin in the perpendicular one. This toy stands in for what every encoder in the project produces: a point cloud whose shape carries the meaning.

THE EXPERIMENT · PCA then whitening

PCA finds the cloud's directions of maximum variance (the EIGENVECTORS) and how much variance each carries (the EIGENVALUES). Here the horizontal eigenvalue is large, the vertical one small. Whitening then divides each direction by the square root of its eigenvalue, so every direction has variance 1 · the stretched cloud becomes round.

THE RESULT · the frame was the problem

The cat/dog SEPARATION was in the cloud all along. Whitening did not create it · it removed the stretch that was hiding it. This is exactly the GPT-2 result (C.11): a collapsed text space, unreadable by cosine, is rescued to 102.6% of a contrastively trained encoder by whitening alone. No new information, just a corrected coordinate frame.

THE MEASUREMENT · what changes, what doesn't

Cosine similarity reads the ANGLE between vectors. In the stretched cloud, the wide direction dominates every angle, so cosine mostly measures that one axis. After whitening, all directions count equally, so cosine reads the full shape. We measure this by retrieval R@1 and by mean pair-cosine before and after.

THE CAVEAT · whitening is not free

Whitening does NOT make cats and dogs identical · it changes the coordinates, never which items are distinct; the two clusters stay separate. And it CUTS BOTH WAYS: the uneven variance it removes is noise when READING a known correspondence (retrieval improves) but SIGNAL when FINDING an unknown one (unsupervised matching collapses, C.12). Isotropy helps one task and destroys the other.

Why this example anchors the whole project. Every headline follows the same logic as cats and dogs: the shared structure is present in the raw spaces (shape), but the coordinate FRAME decides whether a simple metric can see it. Convergence is real and local; the frame · not the content · is what training supplies, and equalising it is a trade, not a universal fix.